

lecture_19

September 19, 2026

1 Lecture 19: Sensitivity Analysis — Non-Linear Programming

{note} Lectures 16–18 each computed a KKT point – the optimal primal variables x^* and the Lagrange multipliers λ^* – using a different algorithm (Penalty, Barrier, Interior-Point). All three methods agreed on the same λ^* , but at the time we treated it only as a byproduct needed to certify optimality. In Lecture 8, we saw that for LP, the multipliers extracted from $\mathbf{B}^{-1}$ are far more than a certificate: they are shadow prices that tell a planner exactly how much an extra unit of a scarce resource is worth. The same is true here. This lecture shows that the KKT multiplier λ_i^* is the NLP shadow price – and develops the resource ranging and cost ranging tools needed to know how far that price can be trusted.

Learning Objectives

By the end of this notebook, you will be able to: 1. Interpret the KKT multiplier λ_i^* as the rate of change of the optimal objective value with respect to the i -th constraint's right-hand side (the NLP Sensitivity Theorem). 2. Determine the range of a constraint's right-hand side (resource ranging) and an objective coefficient (cost ranging) over which the current active set — and hence λ^* — remains valid. 3. Distinguish primal feasibility and dual feasibility as the two distinct mechanisms that can bound a sensitivity range in NLP, and use both to make managerial recommendations in transportation contexts.

Prerequisites: NLP Principles (Lecture 12); Gradient Descent, Newton's Method, BFGS (Lectures 13–15); Penalty Method, Barrier Method, Interior-Point Method (Lectures 16–18); LP Sensitivity Analysis (Lecture 8); calculus (partial derivatives, implicit differentiation); basic linear algebra.

Estimated time: 90 minutes (including in-class exercises)

1.1 Optimality Conditions — KKT

Consider the standard-form NLP:

$$\min_x f(x) \quad \text{subject to} \quad g_i(x) \leq 0, \quad i = 1, \dots, m$$

where $f, g_i : \mathbb{R}^n \rightarrow \mathbb{R}$ are continuously differentiable. The Lagrangian is:

$$\mathcal{L}(x, \lambda) = f(x) + \sum_{i=1}^m \lambda_i g_i(x)$$

If x^* is a local minimum and a constraint qualification holds (e.g., LICQ), then there exist multipliers $\lambda^* \in \mathbb{R}^m$ such that the following Karush–Kuhn–Tucker (KKT) conditions are satisfied:

Condition	Statement	Interpretation
Stationarity	$\nabla f(x^*) + \sum_{i=1}^m \lambda_i^* \nabla g_i(x^*) = \mathbf{0}$	Gradient of objective balanced by gradients of active constraints
Primal feasibility	$g_i(x^*) \leq 0 \quad \forall i$	Solution lies within the feasible region
Dual feasibility	$\lambda_i^* \geq 0 \quad \forall i$	Multipliers are non-negative for inequality constraints
Complementary slackness	$\lambda_i^* g_i(x^*) = 0 \quad \forall i$	A constraint is either active ($g_i = 0$) or its multiplier is zero ($\lambda_i = 0$)

For a convex NLP (convex f , convex g_i), the KKT conditions are both necessary and sufficient for global optimality. Lectures 16–18 each computed a point satisfying these four conditions; this lecture asks what else that point can tell us. We now make this concrete using the NHAH parallel-links problem from Lecture 18.

1.1.1 Worked Example: NHAH Parallel Links

Same problem as Lecture 18, Exercise 2. An NHAH corridor study models total system travel time across two parallel links — the Mumbai–Pune Expressway (v_1 , thousand veh/hr) and old NH48 (v_2 , thousand veh/hr):

$$\min_{v_1, v_2} T(v_1, v_2) = 3v_1^2 - 2v_1v_2 + 2v_2^2 - 14v_1 - 12v_2 + 50$$

subject to a combined capacity constraint $v_1 + v_2 \leq b$, $v_1, v_2 \geq 0$, with baseline $b = 5$ (thousand veh/hr).

1.2 Shadow Price

Transforming the NHAH objective function and constraint into Lagrange Function:

$$\mathcal{L}(v_1, v_2, \lambda_1, \mu_1, \mu_2) = 3v_1^2 - 2v_1v_2 + 2v_2^2 - 14v_1 - 12v_2 + 50 + \lambda(v_1 + v_2 - b) - \mu_1v_1 - \mu_2v_2$$

Assuming the active set — capacity binding, both non-negativity bounds slack, thus forcing $\mu_1 = \mu_2 = 0$, and collapsing the Lagrangian to:

$$\mathcal{L}(v_1, v_2, \lambda) = 3v_1^2 - 2v_1v_2 + 2v_2^2 - 14v_1 - 12v_2 + 50 + \lambda(v_1 + v_2 - 5)$$

Stationarity ($\nabla_{v_1, v_2} \mathcal{L} = \mathbf{0}$) gives two linear equations in v_1, v_2, λ :

$$\frac{\partial \mathcal{L}}{\partial v_1} = 6v_1 - 2v_2 - 14 + \lambda = 0, \quad \frac{\partial \mathcal{L}}{\partial v_2} = -2v_1 + 4v_2 - 12 + \lambda = 0$$

Subtracting eliminates λ (its coefficient is 1 in both) and leaves $8v_1 - 6v_2 = 2$, i.e. $4v_1 - 3v_2 = 1$. Paired with the active constraint $v_1 + v_2 = 5$, this is a 2×2 linear system in (v_1, v_2) :

$$\begin{bmatrix} 4 & -3 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} = \begin{bmatrix} 1 \\ 5 \end{bmatrix} \implies v_1^* = \frac{16}{7}, \quad v_2^* = \frac{19}{7}$$

Back-substituting into either stationarity equation recovers the multiplier:

$$\lambda^* = 14 + 2v_2^* - 6v_1^* = 14 + \frac{38}{7} - \frac{96}{7} = \frac{40}{7} \approx 5.714$$

$$v_1^* = \frac{16}{7}, \quad v_2^* = \frac{19}{7}, \quad \lambda^* = \frac{40}{7} \approx 5.714$$

Since $v_1^*, v_2^* > 0$, $\mu_1 = \mu_2 = 0$, justifying the simplified Lagrangian above; the capacity constraint is active ($v_1^* + v_2^* = 5$) and both non-negativity bounds are inactive.

Note, if the capacity constraint is relaxed by one unit, the total system travel time will drop by 5.714 hours.

We will now treat b and the linear cost coefficient on v_1 as parameters and ask how far each can move before this KKT point — and the value of λ^* — stops being valid.

1.2.1 Resource Ranging

Since the capacity constraint is active at the optimum, we can substitute $v_2 = b - v_1$ directly into T and re-derive the unconstrained-in- v_1 minimum as a function of b :

$$T(v_1, b - v_1) = 7v_1^2 - 2v_1(3b + 1) + 2b^2 - 12b + 50$$

Setting $\partial T / \partial v_1 = 0$:

$$v_1^*(b) = \frac{3b + 1}{7}, \quad v_2^*(b) = b - v_1^*(b) = \frac{4b - 1}{7}$$

Substituting back gives the optimal value function in closed form:

$$f^*(b) = \frac{5}{7}b^2 - \frac{90}{7}b + \frac{349}{7}$$

$$\lambda^*(b) = -\frac{df^*}{db} = \frac{90}{7} - \frac{10}{7}b$$

At $b = 5$: $\lambda^*(5) = 90/7 - 50/7 = 40/7$ — matching the hand-computed value exactly.

Validity range. Two distinct mechanisms can break this formula, and both must be checked — this is the key NLP-specific lesson:

1. Primal feasibility of the active-set assumption: $v_1^*(b) \geq 0$ and $v_2^*(b) \geq 0$ must both hold, or the assumed active set (constraint binding, both bounds inactive) is wrong.

$$v_1^*(b) \geq 0 \Rightarrow b \geq -\frac{1}{3} \quad v_2^*(b) \geq 0 \Rightarrow b \geq \frac{1}{4}$$

2. Dual feasibility: $\lambda^*(b) \geq 0$, or the capacity constraint is no longer binding at all (the unconstrained optimum becomes feasible).

$$\lambda^*(b) \geq 0 \Rightarrow \frac{90}{7} - \frac{10}{7}b \geq 0 \Rightarrow b \leq 9$$

The binding lower bound is $v_2^*(b) \geq 0 \Rightarrow b \geq 1/4$, and the binding upper bound is $\lambda^*(b) \geq 0 \Rightarrow b \leq 9$. So:

$$b \in [\frac{1}{4}, 9]$$

Within this range, $\lambda^*(b) = \frac{90}{7} - \frac{10}{7}b$ gives the *exact, locally-instantaneous* value of relaxing capacity by one unit — but note that, unlike LP, this rate itself changes continuously with b rather than staying constant.

Managerial interpretation: As long as combined corridor capacity stays between 0.25 and 9 thousand veh/hr, the optimum continues to split flow between both links with the capacity constraint binding. At $b = 9$, the constraint stops binding altogether — NHAI would have enough combined capacity that travel time is minimized at the unconstrained optimum $(v_1, v_2) = (4, 5)$, and adding further capacity yields zero marginal benefit ($\lambda^* = 0$).

1.2.2 Cost Ranging

Now hold $b = 5$ fixed and ask how far the linear cost coefficient on v_1 (currently -14) can shift before the active set changes. Write the perturbed objective as:

$$T(v_1, v_2; \Delta c) = 3v_1^2 - 2v_1v_2 + 2v_2^2 + (-14 + \Delta c)v_1 - 12v_2 + 50$$

Substituting the active constraint $v_2 = 5 - v_1$ and re-solving $\partial T / \partial v_1 = 0$:

$$v_1^*(\Delta c) = \frac{16}{7} - \frac{\Delta c}{14}, \quad v_2^*(\Delta c) = \frac{19}{7} + \frac{\Delta c}{14}$$

From stationarity, $\lambda(\Delta c) = -[6v_1^*(\Delta c) - 2v_2^*(\Delta c) + (-14 + \Delta c)]$, which simplifies to:

$$\lambda(\Delta c) = \frac{40}{7} - \frac{3\Delta c}{7}$$

Checking both feasibility mechanisms again: - Primal feasibility ($v_1^* \geq 0, v_2^* \geq 0$): $v_1^*(\Delta c) \geq 0 \Rightarrow \Delta c \leq 32$; $v_2^*(\Delta c) \geq 0 \Rightarrow \Delta c \geq -38$. - Dual feasibility ($\lambda(\Delta c) \geq 0$): $\Delta c \leq \frac{40}{3} \approx 13.33$.

The binding bound is dual feasibility, not primal feasibility:

$$\Delta c \in \left[-38, \frac{40}{3}\right], \quad \text{i.e. the coefficient on } v_1 \in \left[-52, -14 + \frac{40}{3}\right] = \left[-52, -\frac{2}{3}\right]$$

{caution} A student instinctively checks only $v_1^*, v_2^* \geq 0$ and would report $\Delta c \in [-38, 32]$ - too wide. The active set does change at $\Delta c = 40/3$, well before v_1 or v_2 would have gone negative: at that point λ hits zero and the capacity constraint stops binding altogether. Always check dual feasibility ($\lambda \geq 0$) alongside primal feasibility when ranging in NLP - in LP this distinction is invisible because the simplex tableau already enforces both simultaneously, but in NLP the two checks must be made separately.

Managerial interpretation: The current flow split $(v_1^*, v_2^*) = (16/7, 19/7)$ and the value $\lambda^* = 40/7$ remain valid as long as the marginal cost coefficient on the expressway stays between -52 and $-2/3$.

1.3 In-Class Exercises

1.3.1 Exercise 1

Same problem as Lecture 18, In-Class Exercise 1. MTC Chennai is optimizing a demand-responsive feeder service on the Tambaram–Chromepet corridor. The daily operating cost depends on fleet size n (vehicles) and scheduled headway h (minutes):

$$C(n, h) = (n - 5)^2 + 2(h - 3)^2$$

subject to a depot resource constraint $n + h \leq b$, with baseline $b = 7$. Lecture 18 verified by hand that the KKT point at $b = 7$ is $n^* = 13/3, h^* = 8/3, \lambda^* = 4/3$.

1. Substitute the active constraint $h = b - n$ into C , and re-derive $n^*(b)$, $h^*(b)$, and $C^*(b)$ as closed-form functions of b . Confirm $C^*(7)$ matches the Lecture 18 value.
2. Differentiate to obtain $\lambda^*(b) = -dC^*/db$, and confirm $\lambda^*(7) = 4/3$.
3. Find the resource-ranging interval for b , checking both primal feasibility ($n^*(b) \geq 0, h^*(b) \geq 0$) and dual feasibility ($\lambda^*(b) \geq 0$). Which bound binds on each side?
4. Now hold $b = 7$ fixed and let the fleet-size target shift: replace $(n - 5)^2$ with $(n - 5 - \Delta c)^2$. Re-derive $n^*(\Delta c)$, $h^*(\Delta c)$, and $\lambda(\Delta c)$. Find the valid range of Δc .
5. The MTC depot manager asks: “If I free up one more unit of combined depot capacity, exactly how much does today’s operating cost fall?” Answer using $\lambda^*(7)$, then verify numerically using `scipy.optimize.minimize` at $b = 7$ and $b = 8$.

1.3.2 Exercise 2

Same problem as Lecture 18, Take-Away Exercise 1. Delhivery is minimizing the daily operating cost at its Chennai cross-dock facility, with cost depending on daily throughput p (tonnes/day) and

staffing level w (workers):

$$C(p, w) = 2p^2 - 4pw + 3w^2 - 12p - 6w + 50$$

subject to a combined budget-shift constraint $p + 2w \leq b$, baseline $b = 14$. The Lecture 18 KKT point is $p^* = 116/19$, $w^* = 75/19$, $\lambda^* = 64/19$.

1. Substitute the active constraint $w = (b - p)/2$ into C , and derive $p^*(b)$ and $C^*(b)$ in closed form.
 2. Derive $\lambda^*(b)$ and confirm it matches $64/19$ at $b = 14$.
 3. Find the resource-ranging interval for b . (*Hint: as b decreases, which variable hits zero first — p^* or w^* ?*)
 4. The Delhivery facility manager originally asked (Lecture 18) what happens if the budget increases from 14 to 15. Answer this precisely now using $\lambda^*(b)$ rather than the single-point estimate from Lecture 18, and explain why the two answers differ slightly.
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1.4 Take-Away Exercises

For each problem below: derive $f^*(\theta)$ in closed form along the active constraint, derive $\lambda^*(\theta) = -df^*/d\theta$, find the resource- and cost-ranging intervals (checking both primal and dual feasibility), and state the managerial interpretation.

1.4.1 Exercise 1

Same problem as Lecture 18, Take-Away Exercise 2. BMTC is optimizing its demand-responsive feeder service in Bengaluru. The total daily operating cost depends on fleet size n (vehicles) and headway h (minutes):

$$C(n, h) = 2n^2 + 3h^2 + 2nh - 20n - 30h + 100$$

subject to a depot capacity constraint $n \leq b$, baseline $b = 2$. The Lecture 18 KKT point is $n^* = 2$, $h^* = 13/3$, $\nu^* = 10/3$.

1. For fixed $n = b$, minimize over h to obtain $h^*(b)$, then derive $C^*(b)$ and $\nu^*(b) = -dC^*/db$ in closed form. Confirm $\nu^*(2) = 10/3$.
2. Find the resource-ranging interval for b . (*Note: h is unconstrained here, so only the constraint $n \leq b$ and dual feasibility $\nu \geq 0$ can bind — there is no lower primal-feasibility bound on n from this constraint alone. Does the problem have an implicit $n \geq 0$ bound to consider?*)
3. BMTC's depot engineer reports the depot can be expanded to allow $n \leq 3$. Using $\nu^*(b)$, predict the new operating cost, then verify by re-solving the NLP numerically.

1.4.2 Exercise 2

A Kochi container freight station is minimizing daily handling cost as a function of throughput at two berths, x_1 and x_2 (TEU/hr):

$$C(x_1, x_2) = 4x_1^2 - 2x_1x_2 + 3x_2^2 - 20x_1 - 16x_2 + 80$$

subject to a combined crane-hour constraint $x_1 + x_2 \leq b$, baseline $b = 6$, and $x_1, x_2 \geq 0$.

1. Find the KKT point at $b = 6$ by hand: substitute the active constraint, solve the resulting one-variable stationarity condition, and recover λ^* .
 2. Derive $C^*(b)$ and $\lambda^*(b)$ in closed form, and find the resource-ranging interval for b .
 3. Now fix $b = 6$ and perturb the linear coefficient on x_1 (currently -20) by Δc . Derive $\lambda(\Delta c)$ and find the cost-ranging interval. Which feasibility mechanism binds?
 4. Implement a Python function `sensitivity_report(b)` that takes a capacity value, solves the NLP via `scipy.optimize.minimize`, and returns $(x_1^*, x_2^*, C^*, \hat{\lambda})$, where $\hat{\lambda}$ is recovered numerically as the constraint's Lagrange multiplier (`res.get('multipliers')`) or via the KKT residual, as in Lecture 18). Compare $\hat{\lambda}$ against your closed-form $\lambda^*(b)$ across $b \in \{4, 5, 6, 7, 8\}$.
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1.5 Circling Back

- Lecture 8 (LP Sensitivity Analysis): The structure of this lecture mirrors Lecture 8 closely — shadow prices, resource ranging, and cost ranging all reappear. The key difference is *where the multiplier comes from* (the simplex basis $\mathbf{B}^{-1}$ in LP vs. the KKT stationarity condition in NLP) and *how ranging is computed* (closed-form vertex enumeration in LP vs. local active-set/second-order analysis in NLP).
- Lectures 16–18 (Penalty, Barrier, Interior-Point Methods): Each method converges to the same KKT point and the same λ^* — this lecture is the natural sequel, extracting managerial value from that shared output rather than treating λ^* as a stopping-criterion artefact.
- Lecture 12 (NLP Principles): The Taxonomy table previewed that “Lagrange multipliers measure constraint sensitivity” well before Lectures 16–18 introduced the machinery to compute them — this lecture is where that early preview is finally made precise and operational.

1.6 Moving Forward

- Lecture 20 (NLP Duality): The Sensitivity Theorem used here ($\lambda_i^* = -\partial f^*/\partial b_i$) is a special case of a much more general result connecting the primal NLP to its dual problem — exactly as LP shadow prices in Lecture 9 turned out to be the optimal dual variables. NLP Duality formalizes this connection and shows when strong duality (zero optimality gap) holds.
 - Lecture 21 (TE Problems III — Facility Location): Facility location problems will use sensitivity analysis to answer questions like “how much would relaxing a demand-coverage requirement save?” — directly applying the resource-ranging machinery developed here.
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1.7 Further Reading

- Nocedal, J. and Wright, S.J. (2006). *Numerical Optimization* (2nd ed.). Springer — Chapter 12 (theory of constrained optimization, sensitivity and active-set behaviour).
- Bazaraa, M.S., Sherali, H.D., and Shetty, C.M. (2006). *Nonlinear Programming: Theory and Algorithms* (3rd ed.). Wiley — Chapter 4 (Lagrangian duality and sensitivity for nonlinear programs).
- Fiacco, A.V. (1983). *Introduction to Sensitivity and Stability Analysis in Nonlinear Programming*. Academic Press — the canonical reference on parametric NLP sensitivity.

- SciPy documentation: `scipy.optimize.minimize` with `method='SLSQP'` or `'trust-constr'` returns Lagrange multiplier estimates via `res.v` / KKT residuals — docs.scipy.org